

CLINICAL FINDINGS

Childhood is marked by periods of significant and rapid growth and development, and nutritional deprivation often manifests as alterations in these patterns. **Failure to thrive**, a key finding in patients with **protein-energy malnutrition (PEM)**, may appear first as **wasting** (poor weight gain or weight loss), and eventually progress to **stunting** (decreased linear growth).

Recommended markers for malnutrition include: - Body weight and length/height relative to age

- Body mass index (BMI)
- Triceps or mid-upper arm circumference
- Skin and hair characteristics

Chronically malnourished children may also exhibit: - Suboptimal neurodevelopmental outcomes

- Increased susceptibility to infections
- Elevated risk of mortality

Marasmus

Marasmus typically affects infants younger than 1 year of age. Physical findings include: - **Dry, thin, loose, wrinkled skin** due to loss of subcutaneous fat, resulting in an emaciated appearance

- **Slowed hair growth**, easy hair loss, leading to thin, fine, brittle hair and **alopecia**
- Presence of **increased lanugo hair**
- **Nail fissuring** and impaired nail growth

As the body utilizes all available energy stores: - Loss of **subcutaneous fat** and **muscle mass** occurs

- Loss of **buccal fat pads** leads to a characteristic aged or "wizened" facial appearance, historically termed "**monkey facies**"
- **Perianal fat loss** may result in **rectal prolapse**
- **Abdominal muscle hypotonia** can cause **abdominal distention**

Gastrointestinal symptoms include: - Alternating **constipation** and **diarrhea**, with or without concurrent gastrointestinal infection
- **Angular cheilitis** and mucous membrane changes

Additional clinical features: - **Decreased resting body temperature**
- **Bradycardia**

Kwashiorkor

Kwashiorkor, also known as **edematous** or "**wet**" **PEM**, is characterized by: - **Failure to thrive**

- **Edema** (typically starting in the lower extremities and progressing to generalized swelling)
- Typically affects children between **6 months and 5 years of age**

Behavioral changes: - Initial **irritability**, progressing to **lethargy** and **apathy**

Dermatologic manifestations: - "**Flaky paint**" or "**crazy pavement**" **dermatosis**: areas of hyperpigmentation with desquamation and fissuring, resembling cracked paint or pavement

- **Hyperpigmentation** on extensor surfaces of arms and legs

Hair changes: - Early stage: hair develops a **red or reddish tint**

- Later stage: **pigment loss** leads to **light, gray-white hair**

Hypoproteinemia, edema, and diarrhea are prominent. Patients are unable to synthesize lipoproteins, leading to **fat accumulation in the liver** (hepatic steatosis). Critically, **immune proteins are not produced**, resulting in **increased susceptibility to opportunistic infections and septicemia**, which are major causes of mortality.

The exact reason why some children develop **marasmus** versus **kwashiorkor** under conditions of starvation remains controversial, despite the conceptual frameworks of adapted and nonadapted starvation.